A Sharp Norm Bound for Weighted Up-Operators on the Boolean Lattice

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Abstract

For $a = (a_1, \dots, a_m) \in \mathbb{R}^m$ and $1 \le k \le m$, let $U_k(a) : \ell^2\binom{[m]}{k} \to \ell^2\binom{[m]}{k-1}$ be the weighted up-operator (the weighted $(k, k-1)$ inclusion matrix),

$$(U_k(a)f)(S) = \sum_{i \notin S} a_i \, f(S \cup \{i\}), \qquad |S| = k-1.$$

Writing $\|a\|_2^2 = \sum_{i=1}^m a_i^2$, the sharp bound is

$$\|U_k(a)\|^2 \;\le\; \frac{k(m-k+1)}{m}\,\|a\|_2^2 \;=\; \frac{\|U_k(\mathbf{1})\|^2}{m}\,\|a\|_2^2.$$

In words: among weight vectors of a fixed Euclidean norm, the *equal-magnitude* vectors $a = (\pm c, \dots, \pm c)$ maximize the operator norm, and for interior k and $a \neq 0$ they are the only maximizers. The equal-weight identity $\|U_k(\mathbf{1})\|^2 = k(m-k+1)$ is classical and follows from the Johnson-scheme block diagonalization of the Boolean lattice; generic weights break that symmetry, so a different argument is needed. The proof below is a coordinate-deletion induction on m whose scalar closure defect factors as a perfect square. At the endpoints $k = 1$ and $k = m$ equality holds for every a.

1 Introduction

For integers $m \ge 2$ and $0 \le j \le m$ let $\binom{[m]}{j}$ denote the family of j-element subsets of $[m] = \{1, \dots, m\}$, and let $\ell^2\binom{[m]}{j}$ be the Euclidean space with orthonormal basis $\{e_S : |S| = j\}$. For a weight vector $a \in \mathbb{R}^m$ and $1 \le k \le m$, the *weighted up-operator*

$$U_k(a) : \ell^2\binom{[m]}{k} \to \ell^2\binom{[m]}{k-1}, \qquad (U_k(a)f)(S) = \sum_{i \notin S} a_i \, f(S \cup \{i\})$$

is the $(k, k-1)$ inclusion matrix of the Boolean lattice, with the i-th coordinate scaled by the weight a_i. When $a = \mathbf{1} = (1, \dots, 1)$ this is the ordinary up-operator of the Boolean lattice; the weight vector *deforms* that operator while leaving the underlying lattice fixed.

The main result of this note is the sharp inequality

$$\boxed{\|U_k(a)\|^2 \;\le\; \frac{\|U_k(\mathbf{1})\|^2}{m}\,\|a\|_2^2 \;=\; \frac{k(m-k+1)}{m}\sum_{i=1}^m a_i^2} \tag{1}$$

(Theorem 1), and the exact characterization of its equality case. Its content, stated in words, is that *the unweighted operator is the worst case*: no way of distributing a fixed Euclidean budget of weight across the m coordinates can make the operator norm exceed what the flat vector achieves. Equivalently, for fixed $\|a\|_2$ the operator norm is maximized exactly by the equal-magnitude weights, $|a_1| = \cdots = |a_m|$, and — for interior levels and $a \neq 0$ — by nothing else.

The inequality is sharp because $U_k(\mathbf{1})$ attains the bound. What makes the statement non-obvious is that it is *not* a corollary of the classical spectral theory of the Boolean lattice. For $a = \mathbf{1}$ the operators U_k and U_k^{T} intertwine with the levelwise action of the symmetric group, so $U_k U_k^{\mathsf{T}}$ lies in the Bose–Mesner (Johnson) association scheme and its spectrum is obtained by a standard simultaneous block diagonalization [1, 2]. A generic weight vector destroys exactly that symmetry: the weighted operators no longer lie in the adjacency algebra of the Johnson scheme, so the equal-weight decomposition says nothing about $\|U_k(a)\|$ whenever a is not flat. The inequality nevertheless survives with the *same* optimal constant $C(m,k) = k(m-k+1)/m$, and the purpose of the proof is to explain why.

The mechanism is a coordinate-deletion induction. Deleting a coordinate splits $U_k(a)$ into an unweighted block acting on $[m-1]$, a single weighted block, and the diagonal weight a_m ((3)); the operator norm of the resulting 2×2 block operator is then controlled by a block-norm lemma, and induction on m carries the constant $C(m,k)$ through unchanged. The induction closes through an exact scalar inequality whose defect is a perfect square ((7)). That square is the only place where the specific constant $k(m-k+1)/m$ is pinned down: it is not imposed, it falls out.

We record what the theorem does and does not say. It concerns a *single* weighted level operator in ℓ^2-operator norm; it makes no statement about the full section operator $\sum_i a_i \sigma_i^-$, whose norm is $\max_k \|U_k(a)\|$, nor about ℓ^p norms for $p \neq 2$. We make no claim of an application, and we make no claim that the deletion induction is the only possible route to (1).

2 Statement of the result

For $a \in \mathbb{R}^m$ the operator $U_k(a)$ is, in the subset basis, the weighted $(k, k-1)$ inclusion matrix: its (S,T) entry is a_i if $T = S \cup \{i\}$ and 0 otherwise. For $a = \mathbf{1}$ its spectral theory is classical; in particular $\|U_k(\mathbf{1})\|^2 = k(m-k+1)$ [1, 2].

Theorem 1. *For all $m \geq 2$, $1 \leq k \leq m$, and $a \in \mathbb{R}^m$,*

$$\|U_k(a)\|^2 \;\leq\; \frac{k(m-k+1)}{m} \sum_{i=1}^{m} a_i^2.$$

The constant $C(m,k) = \frac{k(m-k+1)}{m}$ is best possible. Moreover:

(i) for $k = 1$ and $k = m$ equality holds for every a;

(ii) for $1 < k < m$ and $a \neq 0$, equality holds if and only if $|a_1| = \cdots = |a_m|$; in particular no weight vanishes.

The equal-weight case gives $\|U_k(\mathbf{1})\|^2 = k(m-k+1) = C(m,k) \cdot m$, so (1) asserts that arbitrary coordinate weights, measured in ℓ^2, can never exceed the rescaled classical constant.

3 Proof

3.1 Sign reduction and the two Gram matrices

For $\sigma \in \{\pm 1\}^m$ and $0 \le j \le m$ let $D_{\sigma,j}$ be the diagonal orthogonal operator on $\ell^2\binom{[m]}{j}$ given by $(D_{\sigma,j}e)_T = \big(\prod_{i\in T}\sigma_i\big)e_T$. Then, for $1 \le k \le m$,

$$U_k(\sigma\cdot a) \;=\; D_{\sigma,k-1}\,U_k(a)\,D_{\sigma,k}. \tag{2}$$

Indeed, for $T = S \cup \{i\}$ the (S,T) entry of the right side is $\big(\prod_{j\in S}\sigma_j\big)a_i\big(\prod_{l\in T}\sigma_l\big) = \sigma_i a_i$, the (S,T) entry of the left side; all other entries agree as zero. Both factors in (2) are orthogonal, so $\|U_k(a)\| = \|U_k(|a|)\|$. We assume $a \ge 0$ from now on; the general case follows, and the equality classification inherits the reduction.

We shall use two Gram matrices of $U_k(a)$. Since $U_k(a)$ is a linear map between finite-dimensional Euclidean spaces, the *two* Gram matrices

$$G_k(a) := U_k(a)^{\mathsf T}U_k(a), \qquad H_k(a) := U_k(a)U_k(a)^{\mathsf T}$$

have the same nonzero eigenvalues — both are the squares of the singular values of $U_k(a)$ — and hence the same largest eigenvalue, equal by definition to $\|U_k(a)\|^2$. We use $G_k(a)$ where the levelwise conjugation (2) is the natural object (it gives $G_k(\sigma\cdot a) = D_{\sigma,k}G_k(a)D_{\sigma,k}$, so the whole spectrum of G_k is preserved), and $H_k(a)$ in the deletion induction, where the block structure (4) is the natural object.

3.2 Block decomposition by deleting a coordinate

Write $a = (a', a_m)$ with $a' \in \mathbb{R}^{m-1}$, and set $A = U_k(a')$, $B = U_{k-1}(a')$. Order the rows of $U_k(a)$ by $(k-1)$-sets not containing m first and those containing m second, and the columns likewise by k-sets. Since the (S,T) entry is nonzero only when $T = S \cup \{i\}$,

$$U_k(a) \;=\; \begin{pmatrix} A & a_m I \\ 0 & B \end{pmatrix}, \tag{3}$$

where I is the $\binom{m-1}{k-1} \times \binom{m-1}{k-1}$ identity indexed by the bijection $S \leftrightarrow S\cup\{m\}$ between the $(k-1)$-sets of $[m-1]$ and the $(k-1)$-sets of $[m]$ containing m; A has shape $\binom{m-1}{k-1} \times \binom{m-1}{k}$ and B has shape $\binom{m-1}{k-2} \times \binom{m-1}{k-1}$. Consequently

$$H_k(a) = U_k(a)\,U_k(a)^{\mathsf T} \;=\; \begin{pmatrix} AA^{\mathsf T} + a_m^2 I & a_m B^{\mathsf T} \\ a_m B & BB^{\mathsf T} \end{pmatrix}. \tag{4}$$

The off-diagonal blocks have shapes $\binom{m-1}{k-1} \times \binom{m-1}{k-2}$ and $\binom{m-1}{k-2} \times \binom{m-1}{k-1}$, matching the block row and column sizes. (At the boundary level $k = 2$ the lower block has $\binom{m-1}{0} = 1$ row; the endpoint levels $k = 1$ and $k = m$ are handled separately below and never require (3).)

3.3 A block-norm lemma

Lemma 2. *If*

$$M = \begin{pmatrix} P & Q \\ Q^{\mathsf T} & R \end{pmatrix}$$

is real symmetric with P, R square symmetric, then

$$\lambda_{\max}(M) \;\le\; \lambda_{\max}\begin{pmatrix} \lambda_{\max}(P) & s_1(Q) \\ s_1(Q) & \lambda_{\max}(R) \end{pmatrix},$$

where s_1 is the largest singular value.

Proof. For a unit vector $x = (u, v)$, writing $\alpha = |u|$, $\beta = |v|$,

$$x^{\mathsf T} M x \le \lambda_{\max}(P)\,\alpha^2 + \lambda_{\max}(R)\,\beta^2 + 2\,s_1(Q)\,\alpha\beta = y^{\mathsf T} S y, \qquad y = (\alpha, \beta),\ |y| = 1,$$

by Cauchy–Schwarz applied to $u^{\mathsf T} Q v$. Maximizing over unit y gives $\lambda_{\max}(S)$. □

3.4 The induction

We prove the theorem, simultaneously in k, by induction on m, carrying the equality clause of Theorem 1(ii). The endpoints are exact: $U_1(a)$ is the row vector $(a_1, \dots, a_m)$ and $U_m(a)$ is a column vector, so $\|U_1(a)\|^2 = \|U_m(a)\|^2 = |a|^2 = C(m,1)|a|^2 = C(m,m)|a|^2$.

Let $1 < k < m$ and $a \neq 0$. If $a' = 0$, then (3) gives

$$U_k(a) = \begin{pmatrix} 0 & a_m I \\ 0 & 0 \end{pmatrix},$$

which is a_m times a partial isometry: its only nonzero block is the identity carrying the k-sets of $[m]$ containing m isometrically onto the $(k-1)$-sets containing m, so $\|U_k(a)\| = a_m$ and $\|U_k(a)\|^2 = a_m^2 < C(m,k)\,a_m^2$ because $k(m-k+1) - m = (k-1)(m-k) > 0$; the inequality is strict in this case.

Assume then $s := |a'|^2 > 0$ and set $t = a_m^2/s$, $\alpha^2 = \|U_k(a')\|^2$, $\beta^2 = \|U_{k-1}(a')\|^2$. Applying Lemma 2 to (4) with $P = AA^{\mathsf T} + a_m^2 I$, $R = BB^{\mathsf T}$, $Q = a_m B^{\mathsf T}$, and using $s_1(a_m B^{\mathsf T}) = a_m \beta$,

$$\|U_k(a)\|^2 \;\le\; \lambda_{\max}\begin{pmatrix} \alpha^2 + a_m^2 & a_m\beta \\ a_m\beta & \beta^2 \end{pmatrix}. \tag{5}$$

By the induction hypothesis applied on $m-1$, $\alpha^2 \le A_0\, s$ and $\beta^2 \le B_0\, s$ with $A_0 = C(m-1,k) = \frac{k(m-k)}{m-1}$ and $B_0 = C(m-1,k-1) = \frac{(k-1)(m-k+1)}{m-1}$. For a symmetric matrix $\begin{pmatrix} p & o \\ o & q \end{pmatrix}$ the largest eigenvalue $\frac{1}{2}\big(p + q + \sqrt{(p-q)^2 + 4o^2}\big)$ is nondecreasing in p, in q, and in $|o|$ (partial derivatives $\frac{1}{2}\big[1 \pm (p-q)/\sqrt{(p-q)^2+4o^2}\big] \ge 0$, and monotone in $|o|$); hence, from (5),

$$\|U_k(a)\|^2 \;\le\; s \cdot \lambda_{\max}\begin{pmatrix} A_0 + t & \sqrt{tB_0} \\ \sqrt{tB_0} & B_0 \end{pmatrix}. \tag{6}$$

3.5 Scalar closure

Lemma 3. *For $m \ge 3$, $1 \le k \le m$, $t \ge 0$, with A_0, B_0 as above and $C = C(m,k) = \frac{k(m-k+1)}{m}$,*

$$\lambda_{\max}\begin{pmatrix} A_0 + t & \sqrt{tB_0} \\ \sqrt{tB_0} & B_0 \end{pmatrix} \;\le\; C(1+t),$$

and for interior k equality holds if and only if $t = \frac{1}{m-1}$.

Proof. The claim is $\sqrt{(A_0 - B_0 + t)^2 + 4B_0 t} \le X(t)$ with $X(t) = 2C(1+t) - A_0 - B_0 - t$. Now X is affine with slope $2C - 1 = \frac{2k(m-k+1)-m}{m} > 0$ and intercept $X(0) = 2C - A_0 - B_0 = \frac{k(k-1)+(m-k)(m-k+1)}{m(m-1)} > 0$; hence $X(t) > 0$ for $t \ge 0$ and squaring is legitimate. The squared defect factors exactly:

$$X(t)^2 - \big[(A_0 - B_0 + t)^2 + 4B_0 t\big] = \frac{4k(k-1)(m-k)(m-k+1)}{m^2(m-1)^2}\big((m-1)t - 1\big)^2 \;\ge\; 0, \quad (7)$$

a polynomial identity in m, k, t. For interior k the prefactor is positive, so equality holds iff $t = \frac{1}{m-1}$. □

Combining (6) with Lemma 3, $\|U_k(a)\|^2 \le s\,C(1+t) = C(m,k)(s + a_m^2) = C(m,k)|a|^2$, completing the induction. Sharpness: $G_k(\mathbf{1})\mathbf{1} = k(m-k+1)\mathbf{1}$, so equality is attained at $a = \mathbf{1}$ in every level. □

4 Equality cases

Throughout this section $1 < k < m$ and $a \neq 0$. The "if" direction: if $|a_1| = \cdots = |a_m| = c$, the sign reduction (2) gives $\|U_k(a)\| = \|U_k(c\mathbf{1})\| = c\sqrt{k(m-k+1)} = C(m,k)^{1/2}|a|$.

For the only-if direction, suppose equality holds in Theorem 1. The chain of §3 is a chain of upper bounds

$$\|U_k(a)\|^2 \;\le\; (\text{corner at } \alpha^2, \beta^2) \;\le\; (\text{corner at } A_0 s, B_0 s) \;\le\; C(m,k)(1+t)s, \quad (8)$$

whose first link is Lemma 2 applied to (4), whose second replaces (α^2, β^2) by $(A_0 s, B_0 s)$, and whose third is Lemma 3. Equality in the last term of (8) forces equality in each link.

Third link. By Lemma 3, equality in (7) forces $t = \frac{1}{m-1}$; in particular $a_m \neq 0$.

Second link. Both corners in this comparison are 2×2 matrices $\begin{pmatrix} p & o \\ o & q \end{pmatrix}$ with a *nonnegative* off-diagonal entry: (5) is the corner at $(p, q, o) = (\alpha^2 + a_m^2,\ \beta^2,\ a_m\beta)$, and s times the matrix of (6) is the corner at $(A_0 + t,\ B_0,\ \sqrt{tB_0})$; the link bounds the first by the second through the three monotonicities recorded before (6). The third link already forces $t = \frac{1}{m-1} > 0$, hence $a_m \neq 0$; and $\beta \neq 0$, for $\beta = 0$ means $U_{k-1}(a') = 0$, i.e. $a' = 0$ (each a'_i appears as an entry of $U_{k-1}(a')$), which is the strict case handled above. Thus both off-diagonal entries are strictly positive, and for $o > 0$ the partial derivatives $\frac{1}{2}\big[1 \pm (p-q)/\sqrt{(p-q)^2 + 4o^2}\big]$ are strictly positive because $\sqrt{(p-q)^2 + 4o^2} > |p - q|$ — the largest eigenvalue is strictly increasing in p and in q at fixed $o > 0$. Equality in the second link therefore forces equality in the p- and q-steps, i.e. $\alpha^2 = A_0 s$ and $\beta^2 = B_0 s$.

Lifting. The identities $\alpha^2 = C(m-1,k)|a'|^2$ and $\beta^2 = C(m-1,k-1)|a'|^2$ are equalities in the induction hypothesis at $(m-1,k)$ and $(m-1,k-1)$. If $2 \le k \le m-2$ the pair $(m-1,k)$ is interior and the strengthened induction hypothesis applied through the first identity gives $|a_1| = \cdots = |a_{m-1}|$; this includes $k = 2$, since for $m \ge 4$ the pair $(m-1, 2)$ is interior. If $k = m-1$ and $m \ge 4$, the pair $(m-1,k)$ is an endpoint and the first identity carries no information, but $(m-1, m-2)$ is interior and the same conclusion follows from the second identity. Then $t = \frac{1}{m-1}$ gives $a_m^2 = s/(m-1)$, the same common value, so $|a_1| = \cdots = |a_m|$, and no weight vanishes.

Base case $m = 3$, $k = 2$. Here $G_2(a)$ is the 3×3 matrix indexed by $\{1,2\},\{1,3\},\{2,3\}$:

$$G_2(a) = \begin{pmatrix} a_1^2 + a_2^2 & a_2 a_3 & a_1 a_3 \\ a_2 a_3 & a_1^2 + a_3^2 & a_1 a_2 \\ a_1 a_3 & a_1 a_2 & a_2^2 + a_3^2 \end{pmatrix}, \qquad s = a_1^2 + a_2^2 + a_3^2,$$

and $C(3,2) = \frac{4}{3}$. The certificate $M = \frac{4}{3} s\, I - G_2(a)$ is positive semidefinite: its 1×1 principal minors are $\frac{1}{3}(a_1^2 + a_2^2 + 4a_3^2)$, $\frac{1}{3}(a_1^2 + 4a_2^2 + a_3^2)$, $\frac{1}{3}(4a_1^2 + a_2^2 + a_3^2)$; its 2×2 principal minors are $\frac{s}{9}(a_1^2 + 4a_2^2 + 4a_3^2)$, $\frac{s}{9}(4a_1^2 + a_2^2 + 4a_3^2)$, $\frac{s}{9}(4a_1^2 + 4a_2^2 + a_3^2)$; and

$$\det M = \frac{4}{27}\big[(x_1 + x_2 + x_3)^3 - 27 x_1 x_2 x_3\big], \qquad x_i = a_i^2,$$

which is ≥ 0 by the AM–GM inequality $(x_1 + x_2 + x_3)^3 \geq 27 x_1 x_2 x_3$ for $x_i \geq 0$, with equality if and only if $x_1 = x_2 = x_3$. (The AM–GM equality condition is the standard one: equality forces $x_1 = x_2 = x_3$; if some $x_i = 0$ the inequality is strict unless all vanish, which is excluded.) Thus $\lambda_{\max}(G_2(a)) \leq \frac{4}{3} s$ with equality iff $a_1^2 = a_2^2 = a_3^2$, and at equality the nullspace of M is $\mathrm{span}\{(1,1,1)\}$. This completes the base case and, with the lifting step above, the classification. □

Remark 4 (relation to spectral methods). For $a = \mathbf{1}$ the operators $\{U_k, U_k^{\mathsf{T}}\}$ interact with the rank-grading of the Boolean lattice in a way that makes the Johnson-scheme diagonalization compute the full spectrum [2, 1]. For generic weights a the weighted operators no longer lie in the adjacency algebra of the Johnson scheme, so that diagonalization does not apply to $U_k(a)$; the induction above replaces it with a scalar closure whose defect (7) is exactly the deviation from the equal-weight ray. We do not claim that the deletion induction is the only route to (1).

Acknowledgments and disclosure

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References

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